Binary Search

1. Source Code (C)

```c
#include <stdio.h>

int main() {

    int n, i, search, f = 0, low, high, mid, a[20];

    printf("\n***Binary Search***\n");

    printf("\nEnter the number of elements: ");
    scanf("%d", &n);

    // Input array elements in ascending order
    printf("Enter %d numbers in ascending order:\n", n);

    for (i = 0; i < n; i++) {
        printf("a[%d] = ", i);
        scanf("%d", &a[i]);
    }

    printf("Enter the search element: ");
    scanf("%d", &search);

    low = 0;
    high = n - 1;

    // Binary search
    while (low <= high) {

        mid = (low + high) / 2;

        if (search < a[mid])
            high = mid - 1;

        else if (search > a[mid])
            low = mid + 1;

        else {
            f = 1;
            printf("Element %d found at index %d.\n", search, mid);
            break;
        }
    }

    if (f == 0)
        printf("Element %d is not present in the array.\n", search);

    return 0;
}
```

2. Program Output

```
***Binary Search***

Enter the number of elements: 4

Enter 4 numbers in ascending order:

a[0] = 23
a[1] = 45
a[2] = 67
```

```
a[3] = 87

Enter the search element: 45

Element 45 found at index 1.


***Binary Search***

Enter the number of elements: 4

Enter 4 numbers in ascending order:

a[0] = 56
a[1] = 78
a[2] = 87
a[3] = 90

Enter the search element: 10

Element 10 is not present in the array.
```